λ-terms are in β-normal form if they contain no redexes.

$$\text{is\_in\_nf}(M) \stackrel{\text{def}}{=} \forall M'.\ M \not\to_\beta M'$$

$$\text{has\_nf}(M) \stackrel{\text{def}}{=} \exists M'.\ M \longrightarrow_\beta^* M' \wedge \text{is\_in\_nf}(M')$$

THEOREM (UNIQUENESS OF β-NORMAL FORMS)

$$\forall M, N_1, N_2.\ M \longrightarrow_\beta^* N_1 \wedge M \longrightarrow_\beta^* N_2 \wedge$$
$$\text{is\_in\_nf}(N_1) \wedge \text{is\_in\_nf}(N_2) \Longrightarrow N_1 =_\alpha N_2$$

PROOF. From Church-Rosser, obtain N, such that $N_1 \longrightarrow_\beta^* N$ and $N_2 \longrightarrow_\beta^* N$. However, since N_1 and N_2 are in normal form, we have that $N_1 =_\alpha N =_\alpha N_2$.